prefer a pipeline where the raw data from the camera is first transferred to a workstation, after which all the computation is applied. The enthusiasts also use a similar approach, and in fact many camera models ship with subscriptions to image editing and storing solutions.

At some point though, sheer convenience is going to win. Some cameras do have solutions for image stacking, such as HDR or star trails. LiveND by Olympus allows users to tune the exposure and Past Focus by Panasonic functions like the Lytro camera, allowing the focus to be changed after capturing the image. A few other cameras throw in art filters for nostalgia. These are nowhere near the capabilities of a smartphone. It is difficult to find point and shoot, or even prosumer cameras with simple features such as time lapse and panorama. The British Startup Photograph AI announced an AI Powered interchangeable lens camera called Alice in 2020, with a micro four thirds mount on the front, and a universal phone mount on the back. The camera uses a dedicated chip for running the algorithms,

The Obsbot Tail camera follows the action

The Polaroid Snap

which we may be seeing more of in the future considering that limited processing power is what is holding back computational photography techniques from being used in mirrorless cameras.

Sony's new IMX990 and IMX991 SWIR sensors can capture in both the visible light and infrared spectrum, which are particularly helpful for AI driven tasks, and there will be a need to support these advanced image sensors with AI chips. Drone cameras, action cameras, and security cameras can use these capabilities for obstacle avoidance, face tracking, and face recognition. These categories are together driving the need for increased computational capabilities on the device. The Hoop Cam is a home security camera that uses IR for night vision, and AI for sending alerts if motion or sound is detected, The Obsbot Tail is an action camera with a gimbal that uses AI to track action even through obstacles, such as parkour, kids or pets. The "Tiny" variant is a webcam that does the same, but also recognises gestures. In 2018 Google introduced a smarthome camera called Clips, that used AI to automatically record interesting moments. Once consumers get more comfortable living with cameras running all the time, all around them, that is the kind of applications future cameras can be expected to have. The AI capabilities on these cameras can also be used for interfaces. The later GoPro models already have voice interfaces, and the security cameras work with intelligent assistants such as Alexa and Siri, so the way things are headed it looks like the cameras are going to get smarter. Drone cameras and action cameras can be expected to sport LiDAR scanners similar to the setup used in the iPhone 12, as well as "time of flight" imagery from the infrared sensors used to create a depth map of the surroundings, as implemented in the Kinect. We can also expect devices and systems that retrofit existing camera systems and give them smart capabilities. One such ARM powered device, which is at present a project on Kickstarter, is known as Arsenal, and provides all the things that cameras should be doing anyway. It plugs into mirrorless or dSLR cameras, provides the user with wireless controls using a smartphone app, allows for capturing HDR, long exposure, focus stacking, and time lapses. The AI is trained on a bank of images to choose just the perfect setting for the scene in view, and at the same time, prevents the camera from using the settings that are known to produce subpar shots, and can intelligently recognise the subjects and apply the appropriate settings (fast shutter speed for birds). There is really no reason all of these things shouldn't be in the camera already.